

MedAssistAI - Medical Symptom Analysis & Diseases Prediction System

Individual Contribution Report

Field	Details
Name	Koritala Lakshmi Sahitya
Role	Full Stack Developer
Evaluation Phase	Milestone - 2
Status	In Progress – Approx. 80% of Frontend and 70% of Backend Completed

Overview

This report summarizes the work completed during Milestone – 2 of the MedAssist AI project from the perspective of a Full Stack Developer. During this milestone, the primary focus was on developing and integrating the frontend and backend components required to support the Disease Prediction and Risk Assessment module. The work involved building responsive user interfaces, implementing secure backend services, establishing database connectivity, integrating application modules, and enhancing the overall functionality, scalability, and user experience of the healthcare web application.

Work Completed

Frontend Development

- Designed and developed responsive user interfaces for the Disease Prediction and Risk Assessment module.
- Implemented role-based dashboards for Patients, Doctors, Laboratory Staff, Receptionists, Pharmacists, and Administrators.
- Developed interfaces for symptom analysis, disease prediction results, appointments, prescriptions, medical reports, and profile management.
- Improved navigation, responsiveness, and overall user experience.

Backend Development

- Developed RESTful APIs using Flask for secure communication between the frontend, backend, and database.
- Implemented authentication and role-based authorization for different healthcare users.
- Structured backend modules for scalability and maintainability.
- Integrated backend services with the AI prediction module.

Database Integration

- Established centralized database connectivity for managing patient records, appointments, laboratory reports, prescriptions, and healthcare information.
- Ensured secure and synchronized data management across all user roles.

System Integration

- Integrated frontend and backend components for seamless communication.
- Connected multiple healthcare workflows to enable efficient interaction between different user roles.
- Optimized API communication and backend performance.

Additional Features

- Integrated multilingual language support.
- Implemented chatbot functionality.
- Developed the Nearby Hospitals feature using OpenStreetMap.
- Enhanced application accessibility and usability.

Outcomes

- Successfully developed and integrated the frontend and backend infrastructure supporting the Disease Prediction and Risk Assessment module.
- Established secure communication between the user interface, backend services, database, and AI prediction module.
- Implemented centralized data management and role-based access control.
- Improved application scalability, reliability, maintainability, and overall user experience.

Next Steps

- Complete the integration and testing of all application modules.
- Optimize backend performance and API efficiency.
- Perform end-to-end testing across all user roles.
- Prepare the application for deployment and final project demonstration.

Conclusion

The work completed during this milestone established a strong full-stack foundation for the MedAssist AI web application. The successful integration of frontend, backend, and database components has resulted in a secure, scalable, and user-friendly healthcare platform. This foundation enables smooth integration of AI-powered healthcare services and supports the remaining development activities leading to the final deployment of the project.